

HAN YANG

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EDUCATION

B.Sc., Animal Science, **Fastest recipient of a Provincial Government Honor** Sep. 2022 – Jun. 2026
Southwest University (211 & Double First-Class), GPA: 83.6/100 Chongqing, China
• **Core Courses:** Advanced Mathematics (92); Quantitative Genetics (95); Mathematical Modeling (96)

PUBLICATIONS

- **Papers**
H. Yang. ‘Research on Multi-Object Localization of Rocket Debris’. Frontier and progress of Computational Mathematics and Modeling, 2024, no. 6, 1-24. ISSN: 2313-4194.
- **Patents**
Z. Ding, H. Yang, Z. Peng, R. Xing, X. Guo, and T. Yu. Promotional Column (Drama Media). China National Intellectual Property Administration, Design Patent Application No. 2023307625850. Filing Date: Nov. 21, 2023. (Patent Submitted)

SELECTED HONORS

- **Beijing Challenge Group Annual Scholarship (Top 0.25%),** Beijing Challenge Group Nov. 2023
 - **Southwest University First-Class Scholarship (Top 3%),** Southwest University Dec. 2023
 - **Southwest University Academic Science and Technology Award (Top 2%),** Southwest University Dec. 2023
 - **Chongqing Advanced Individual for Innovation Ability Enhancement (Top 0.09%),** Chongqing Municipal Education Commission (Government), **awarded in the shortest time** Jun. 2024
 - **Southwest University Diligent Endeavor Scholarship (Top 2%),** Southwest University Dec. 2024
 - **VTR Bio-Tech Scholarship (Top 0.08%),** Guangdong VTR Bio-Tech Co., Ltd., **Only one undergraduate and one postgraduate recipient in the Faculty** Mar. 2025
- Received over 10 honors in total.

RESEARCH EXPERIENCE

- **ESI-Curvature Modeling of Egg Trajectories: Morphological Effects** Chongqing, China
Independent Research Jun. 2025 – Jun. 2026
Activities: Completed independently as the project leader. Built a standardized image acquisition platform with a 2.87-degree inclined flexible-contact surface, collecting static contour images of 90 egg samples and 270 rolling videos. Extracted 11 static morphological features (egg shape index ESI, asymmetry index AI, Hu moments, etc.) and 3 dynamic indicators (lateral deviation, attitude-angle variance, velocity coefficient of variation) via machine vision, with vision-based ESI achieving only 0.788% mean error vs. manual measurement. Statistical analysis revealed AI as the dominant driver of rolling instability and higher-order contour features as modulators. Fused the dynamic indicators into a fragility index (RSI 0-100) via PCA and K-means clustering, and trained SVM/RF/GBDT/LR classifiers, with SVM selected as the optimal model. Developed and deployed the web application Egg_RSI_App (Streamlit Cloud) for rolling-risk prediction from static images.
Outputs: Provincial-level Innovation and Entrepreneurship Training Program – Excellent completion.
- **Machine Vision for Coupled Egg Dynamics & Intelligent Sorting** Chongqing, China
Leader May 2025 – May 2026
Activities: Led and supervised a four-member team in machine-vision-based egg analysis and sorting theses: developed kinematic time-series extraction (centroid, attitude, velocity, acceleration), coupled static morphology with dynamic rolling for risk prediction via a PCA-based rolling stability index, trained four ML classifiers with SVM optimal (74.07% on 90 eggs, 270 videos), clustered trajectories into three stability classes via GMM, and built multi-species egg classifiers from fused HSV and LBP features (8 species, 713 images; KNN 99.30%).
Outputs: Graduation thesis project of a four-member group, completed successfully.

- **Research on Multi-Object Localization of Rocket Debris**
Independent Research
Activities: Developed a rocket debris localization and landing-point prediction strategy based on sonic-boom signals: converted WGS84 geodetic coordinates to Earth-centered Cartesian coordinates, built an improved tri-sphere and quad-sphere intersection model solved by nonlinear least squares (MATLAB lsqnonlin) to locate single debris with at least four monitoring devices, applied a traversal algorithm over 256 time combinations with feasibility filtering to match four debris pieces to their arrival times, and introduced weighted least squares with Monte Carlo simulation to analyze localization stability under 0.5 s timing errors.
Outputs: Publish the paper titled “Research on Multi-Object Localization of Rocket Debris” .
 - **Various Mathematical Modeling Competitions**
Independent Research
Activities: Independently completed the modeling, computation, and paper writing for mathematical modeling competitions. Notable works include: the 2024 Certified Cup Challenge (finite-element skull model based on the Ogden constitutive model), the 2024 CUMCM (helical coiling dragon simulation via polar-angle iteration and Simpson’s rule), and the 2025 CUMCM (multi-objective UAV smoke-screen jamming simulation optimized by a particle swarm algorithm with dual-plane projection occlusion modeling).
Outputs: Attain first-class awards at both the national and provincial levels.

Chongqing, China

May 2024 – Aug. 2024

Chongqing, China

Mar. 2023 –Jan. 2026
- SELECTED COMPETITIONS

- **The Chinese Mathematics Competitions (CMC)**, Chinese Mathematical Society, **National First Prize**
 - **“Certification Cup” Mathematical Modeling Challenge**, Computational Systems Biology Society of Operations Research Society of China, **National First Prize**
 - **National Higher Education Institutions Mathematics Ability Challenge**, China International Science and Technology Promotion Association, **National First Prize**
 - **China Undergraduate Mathematical Contest in Modeling (CUMCM)**, China Society for Industrial and Applied Mathematics, **Provincial First Prize**

Mar. 2023

Apr. 2024

Jun. 2024

Sep. 2024

Received over **50 awards** across dozens of competitions; mathematical modeling projects were completed independently.
- WORK EXPERIENCE

- **Data Analyst**, Wuhan Zhenye Information Technology Co., Ltd., Wuhan, China
Assisting clients with data analysis and data modeling tasks, mainly using Python and MATLAB.
 - **Research Assistant**, Beijing Challenge Agricultural Technology Co., Ltd., Yancheng, Jiangsu, China
Collected and analyzed sow feed intake and piglet weaning data during an on-site internship in Yancheng.
 - **Operations Manager**, Chongqing Jiawo Biotechnology Co., Ltd., Chongqing, China
Presented a solid-fertilizer greening project at the China Innovation and Entrepreneurship Roadshow.

Aug. 2025 –Aug. 2026

Feb. 2025 –Apr. 2025

Jul. 2023 –Aug. 2023
- EXTRACURRICULAR ACTIVITIES

- **Graduate Funding Allocation Strategy**, Southwest University Graduate School, Chongqing, China
Proposed a weighted total-integral ratio model and a dual-track model, adopted by Southwest University Graduate School.
 - **Experience Sharing**, Postgraduate Entrance Exam Mathematics Competition, Hunan, China
As a freshman, I authored an article for a publication featured on the official National Competition website, achieving **14,000+** views. [Link](#)
 - **Mathematics Competition Course Recording**, Yunding Mathematical Modeling Agency, Beijing, China
Recorded mathematics competition courses; account has **91,000+** followers and videos gained **28,000+** views. [Link](#)

Sep. 2025 –Oct. 2025

Mar. 2023 –May 2023

Jun. 2024 –Aug. 2024
- SKILLS

- **Research Software**: MATLAB, Python, L^AT_EX, LINGO, Streamlit, SPSS, Codex, Hermes, Claude Code.
 - **Languages**: Chinese (Native), English — IELTS Overall 6.0 (L5.5, R7.5, W6.5, S5.0).